Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Java, C/C++, Python, JavaScript, SQL, R
Software Engineering: Data Structures & Algorithms, Distributed Systems, Operating Systems, Database Management Systems (DBMS), Multithreading & Concurrency, Unit Testing & Test-Driven Development, System Design, Design Patterns
Backend & Frameworks: Flask, FastAPI, Node.js, React.js
Databases & Storage: MySQL, SQLite, Redis (NoSQL)
Tools & Platforms: Git, GitHub, GitHub Actions, Docker, VS Code, IntelliJ IDEA

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Analyzed 8,500+ real-world records using Julius AI and LLM-assisted workflows, extracting 15 actionable business insights through data exploration and prompt engineering.
- Developed structured analytical workflows, validated AI-generated outputs, and translated findings into 4 business recommendations presented to 3 mentors.
- Completed an industry-sponsored capstone project, earning certification from VOIS, Vodafone Idea Foundation, and AICTE.

PROJECTS

***RedisLite* |LINK**

- Built RedisLite, a Redis-inspired in-memory key-value database in Python featuring a custom RESP protocol parser, multithreaded TCP server, thread-safe storage engine, and O(1) LRU cache.
- Designed a modular four-layer architecture implementing Write-Ahead Logging (WAL), crash recovery, lazy and active TTL expiration, and concurrent request handling for reliable data persistence.
- Developed 46 automated unit tests, benchmarking 619K SET operations/sec in memory and 2.8K operations/sec with durable persistence while ensuring correctness and reliability.

***API Rate Limiter Library* |LINK**

- Developed a production-ready API rate-limiting library in Python implementing five industry-standard throttling algorithms with Redis integration and reusable middleware for Flask and FastAPI.
- Designed a modular architecture supporting concurrent request handling, storage abstraction, and framework-independent integration for scalable backend applications.
- Built 30+ automated unit tests and published the library to PyPI for public installation and reuse.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Owned the product end-to-end — designed, built, and automated a receipt-processing pipeline using EasyOCR, OpenCV, and NLP-based parsing, extracting merchant, amount, date, and category from diverse formats with confidence scoring, cutting manual entry time by 40%.
- Independently architected and deployed the full stack (FastAPI, React, SQLAlchemy, SQLite, JWT/OAuth authentication), shipping expense analytics, budget tracking, and a live production dashboard hosted on Render and Vercel.

***Docwise* |LINK**

- Owned the product end-to-end — engineered an AI-powered PDF Q&A system using Google Gemini, achieving 95%+ query accuracy and cutting manual document search time by 40%.
- Independently designed the scalability layer — TF-IDF vectorization, chunking, and real-time processing for 200+ page PDFs — plus chat history and session management, driving a 35% lift in user retention.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate |**LINK** 2026
- AWS APAC Solutions Architecture virtual experience program |**LINK** 2025